

A 65-nm Proactive Power Management Technique With Real-Time Machine Learning Engine for Droop Prediction and Mitigation on Microprocessors

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Abstract—A proactive power management (PM) scheme for mitigating dynamic supply droop is proposed with a fully integrated buck converter, an RISC-V CPU core, a real-time machine learning engine (MLE), and a safety droop guardband. The dynamic supply is modeled using an RV32IM ISA CPU and the model was simulated in a customized co-sim environment. The MLE is used to predict the supply droop on a cycle-by-cycle basis for proactive supply regulation from the integrated buck converter through a fast PWM modulation of the power switches. In addition, to deal with the misprediction of MLE and other long-term supply droops that cannot be captured by MLE, a safety droop guardband is implemented to secure the integrity of the supply voltage. The proposed proactive scheme has been implemented on a 65-nm test chip and demonstrated up to 9.9% higher CPU frequency or 9.2% higher power efficiency compared with prior fast digital LDO schemes.

Index Terms—Fully integrated dc–dc converters, machine learning, proactive power management (PM), real-time voltage detection, supply droop mitigation.

I. INTRODUCTION

POWER integrity has become a major challenge in deep CMOS technology where supply voltage has decreased to only 0.6–1.0 V with considerable variation of workload being processed in a system on chip (SoC). While conventional integrated voltage regulators provide certain mitigation to supply events, the approaches engaging regular control loop of power

converters are too slow to provide regulation for fast supply transients, e.g., from sudden changes of workload, resonant droops, or instruction-specific power surges [1], [2], [3].

Adaptive or throttling techniques are widely explored to mitigate timing-margin failures caused by voltage droop. Tunable replica circuits with adaptive voltage scaling (AVS) were incorporated to monitor critical timing margin and dynamically adjust supply voltage [4]. Adaptive clock managements incorporating frequency-locked loop (FLL) [5], instruction-driven phase scaling [6], [7], [8], droop-responsive clock circuits [9], dynamic variation monitor with clock-gating circuits [10], all-digital auto-calibrating adaptive clock distribution [11], and dynamically adaptive frequency with phase-locked loop (PLL) [12], [13], have also been exploited to mitigate the impact of supply droop. However, most adaptive circuits suffer from performance degradation due to lowered frequency or gated clock, while the effectiveness of such techniques is still limited by the response speed. Direct mitigation of timing errors with dual-mode logics [14] or resilient error detection with recovery circuits [15] was also utilized to relax the response-time constraint of droop mitigation, but challenges of reducing the complexity of recovery/correction circuits remain in such resilient designs.

To combat the fast droop events, previously low dropout (LDO) regulators utilizing threshold crossing time to track the workload variation are developed for fast regulation with PI controller [16] or 1Z1P control [17]. An event-based LDO scheme has also been utilized to provide efficient on-demand regulation to the supply events achieving better performance in droop magnitude, output capacitor size, settling time, and current density [18], [19], [20]. As shown in Fig. 1(a), a speed and efficiency tradeoff between LDO and dc–dc converters is observed where LDO offers sub-nanosecond regulation with lower efficiency compared with slower dc–dc converters with higher efficiency [21], [22]. All the prior works took the approach of “reactive” power management (PM) that either requires extra time/voltage margin that degrades the system efficiency or suffers from the tradeoff mentioned above.

Recently, the concept of “proactive” management has drawn significant interest. Critical-path monitors and distributed

Received 2 March 2024; revised 13 June 2024, 7 August 2024, and 11 September 2024; accepted 2 October 2024. Date of publication 5 November 2024; date of current version 29 May 2025. This article was approved by Associate Editor Jaehyook Choi. This work was supported in part by the SRC Task under Grant 2810.039, Grant 3160.0391, and Grant NSF CCF-1846424. (Corresponding author: Xi Chen.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2024.3479273>.

Digital Object Identifier 10.1109/JSSC.2024.3479273

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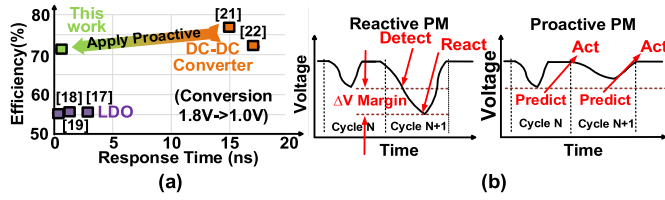


Fig. 1. (a) Speed versus efficiency tradeoff between conventional dc-dc converter and LDO. (b) Concept of conventional reactive and the proposed proactive PM.

power supply estimation engine [24] were used to predict the dangerous droop and then reduce the voltage droop by reducing the instruction fetching rate or lowering the activities in the core logic. A workload-aware throttling with clock-edge skipping technique [25] was developed to exploit the layer-to-layer power variation of neural networks and enhance the performance of an AI chip through throttling higher power layers. A proactive clock throttling technique was also demonstrated on a DSP vector processor where an on-die digital power meter (DPM) and a voltage-clock-gating (VCG) circuit are used to issue clock gating based on prediction of droop violation from microarchitecture events in a few cycles before the execution achieving 10% higher frequency or 5% lower minimal supply [28]. However, the prior throttling approaches incur a performance penalty and do not fundamentally remove the detrimental supply droop. In addition, a comprehensive cycle-accurate design time power meter for high-performance CPU was reported to achieve <1% area/power overhead, without any regulation or feedback control [31].

As modern SoCs often contain fully integrated regulators, proactive PM instead of clock throttling offers an attractive trait as it provides active regulation to combat droop events. As will be described in Section II-B, the on-chip supply voltage of microprocessors not only observes significant variation at cycle-by-cycle bases but also depends on the sequence of program activities as well as the temporal trajectory of supply voltages, leading to much more complicated computational models required for proactive PM. To deal with these challenges, this work demonstrates a proof-of-concept solution for proactive PM for digital SoCs. As shown in Fig. 1(b), the proactive PM predicts the supply events and applies mitigations ahead of time to remove the response speed constraint or the extra voltage margin in conventional schemes. The contributions are summarized as follows.

- 1) A simulation and design platform are developed for machine learning-based prediction of supply droops.
- 2) A real-time machine learning engine (MLE) is designed and implemented on the chip for online droop prediction and mitigation.
- 3) A fast fully integrated buck converter incorporating the MLE and a “safety” guardband for misprediction is also developed to provide proactive management to the high-speed power supply droops.
- 4) A 65-nm test chip has been designed and fabricated to demonstrate the proposed proactive PM technique. The measurement results showed a performance improvement of 9.9% higher CPU frequency or 9.2% higher

power efficiency over prior approaches, demonstrating the promise of proactive PM for a better tradeoff of speed and efficiency.

This article is an extension of the prior conference publication [36]. The rest of this article is organized as follows. Section II introduces the analysis of dynamic power distribution and supply variation for an RISC-V processor. The overall architecture of the proposed proactive PM scheme is presented in Section III. The generation and implementation of MLE and droop guardband are illustrated in Section IV. The simulation and measurement results obtained from the test chip are shown and discussed in Section V. Section VI concludes this work.

II. ANALYSIS AND DROOP PREDICTION ON MICROPROCESSORS

A. Power Analysis and Simulation Flow

In this work, a 32-bit five-stage in-order RISC-V CPU with RV32IM ISA is used as the workload and test vehicle of the proactive PM scheme. Fig. 2 shows the design and simulation environment used in this work for modeling the correlation between the supply events and the program activities of the microprocessor. An RISC-V CPU is implemented through a conventional digital design flow. The GNU toolchain and the Spike simulator are utilized to verify the programs from Mibench [37] and EEMBC. Meanwhile, compiled programs with instructions and cache data are simulated on the RTL design of RISC-V core using Cadence Xcelium. All the events of caches and register file as well as internal signal toggling are recorded for discovering the input features for the droop prediction. Power analysis is performed with the digital backend design of the RISC-V core using Cadence Voltus where current waveforms with power trace information are generated corresponding to the digital signals recorded in Xcelium. An analog mixed-signal environment is used to simulate the CPU current with the power converter to collect runtime supply voltage that contains the droop events corresponding to the benchmark programs. The recorded digital signals and voltage/current waveforms are utilized for droop prediction training and feature selection. To close the loop, the feedback commands issued by the selected model of MLE are further sent back to the power converter in the analog mixed-signal environment to perform a verification on droop mitigation. The model of MLE and feedback circuits of the power converter are further optimized based on the simulated supply waveforms with feedback regulation. A few examples of analysis performed through such multi-hierarchy flow are shown in Fig. 3 and will be further discussed in Section II-B.

B. Power Dependency of Microprocessors

Power analysis on RISC-V CPU with benchmarks from Table I shows an average power of around 150 mW, but cycle-wised average power ranges from ~30 to ~390 mW, as shown in Fig. 4(a). Furthermore, power analysis performed on several testing programs consisting of instructions with specific types shows the correlation between CPU power and instructions. Cache access, a branch that is effectively taken by the program or computations with heavy load, leads to a

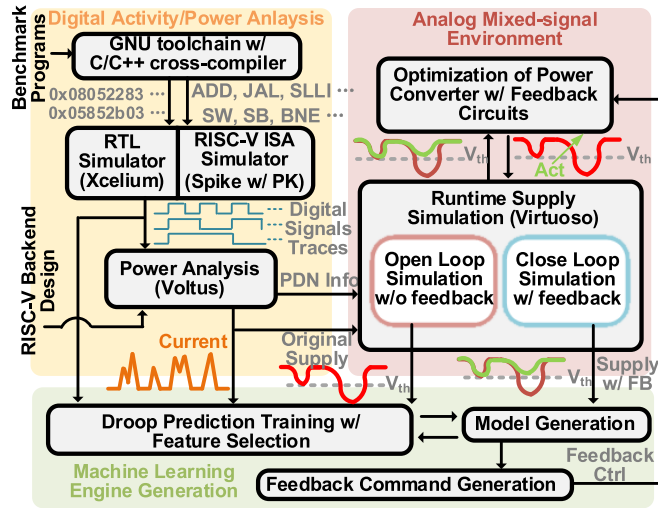


Fig. 2. Customized environment for digital power analysis, analog mixed-signal simulation, and machine learning model generation.

TABLE I
BENCHMARK USED IN THIS ARTICLE

Benchmark Source	Mibench				
Program	StringSearch	BasicMath	FFT	Dijkstra	BitCount
number	BM1	BM2	BM3	BM4	BM5
Benchmark Source	Mibench				EEMBC
Program	Qsort	Ispell	CRC32	Jpeg	CoreMark
number	BM6	BM7	BM8	BM9	BM10

higher power, while logic operation, such as XOR or SLL, consumes less, as shown in Fig. 4(b).

Operations with higher power cause larger supply droop due to the extra current driven through the power trace. Meanwhile, the actual voltage margin is determined not only by the workload but also by the initial/previous values of the supply. Therefore, both CPU activities and supply conditions during the programs need to be considered to predict the droop. However, the complexity of the prediction model needs to be considered because the overhead of the MLE may override its benefit.

As mentioned in Section II-A, Fig. 3 shows the detailed examples and key observations for supply prediction based on the power analysis described in Section II-A, i.e., the digital signal traces of CPU simulated in Xcelium, workload current waveforms from Voltus, and runtime supply waveforms simulated with fully integrated buck converter in Virtuoso. Simulation was based on the program StringSearch and three dimensions of dependency are observed as follows.

- 1) *Instruction/Data Dependency*: As shown in Fig. 4(b), different instruction operations lead to different power levels through the runtime cycles. For instance, the executed branch causes higher current due to a large amount of toggling from flush operation, which results in a voltage change on supply waveforms, while an OR operation leads to a lower power, as shown at the left of Fig. 3.
- 2) *Instruction Sequence Dependency*: As the accumulation effect exists when current consumptions are converted

to voltage droops due to capacitors and inductors in the power supply network (PSN), continuous instructions with heavy load can result in larger and severer undershoots, e.g., a sequence of four store instruction caused an accumulated larger droop as shown at upper right of Fig. 3. On the contrary, continuous light-load instructions can cause the voltage to recover or even result in an overshoot. Moreover, the influence of interleaving heavy- and light-load instructions can cancel each other within a certain period, as shown at the bottom right of Fig. 3.

- 3) *Voltage Dependency*: As previously discussed, supply droop is influenced by both CPU activities and real-time voltage trajectories. In a real chip environment, the supply voltage depends on multiple factors, such as impedance of PSN, decoupling capacitors of the chip, low-frequency oscillation from PCB traces, and power converter control loops, all of which contribute to the transient of supply voltage. As shown in the middle of Fig. 3, the supply waveform has a dynamic varied slow-moving trajectory that impacts whether a certain droop will reach the threshold. As a result, both the current supply voltage and traces of the supply voltages need to be considered for prediction of future supply droops.

Through the discussion above, the detailed instruction-based operations of each cycle as well as its sequency pattern and cycle-wise real-time supply voltages are all required to perform the droop prediction, which will be further discussed in Section IV.

III. OVERVIEW OF THE PROPOSED ARCHITECTURE

A. Proactive Real-Time PM Scheme

Fig. 5 shows the overall chip architecture of the proposed proactive PM scheme. An RISC-V CPU core running at 0.6–1.2 GHz from a nominal supply voltage of 1.0–1.2 V is powered by a fully integrated buck converter, which includes a feedback control loop at 10–50 MHz and a fast PWM generator with PWM signals at 0.6–1 GHz for two-phase interleaved power cores. The fully integrated buck converter utilizes on-chip spiral inductors to convert an IO input voltage of 1.8 V to CPU's supply voltage between 0.6 and 1.4 V. MLE is used to generate the prediction of supply droop and issues feedforward power regulation through direct modulation of the PWM control signals. A guardband module consisting of a long-term guardband (LTG) and a short-term guardband (STG) circuit is used to provide a “safety net” for missed regulations from the MLE. To support guardband operations, a group of fast comparators is used for detecting the real-time voltage level with a latency of less than 200 ps. For testing purposes, a cycle-by-cycle voltage recorder (VR) with 2k depth is implemented to record real-time supply voltage information.

B. Fully Integrated Buck Converter

Fig. 6(a) shows the design of the fully integrated buck converter. Two interleaving power cores are implemented to deliver up to 200-mW output. Each power core consists of

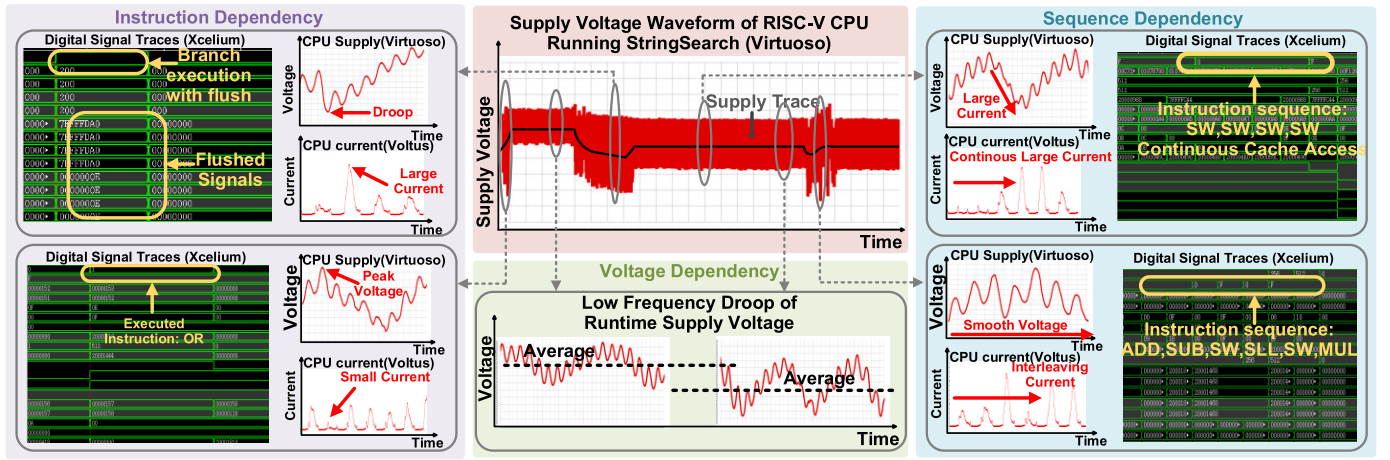


Fig. 3. Examples from simulations showing supply droop dependency on CPU behaviors and real-time voltage.

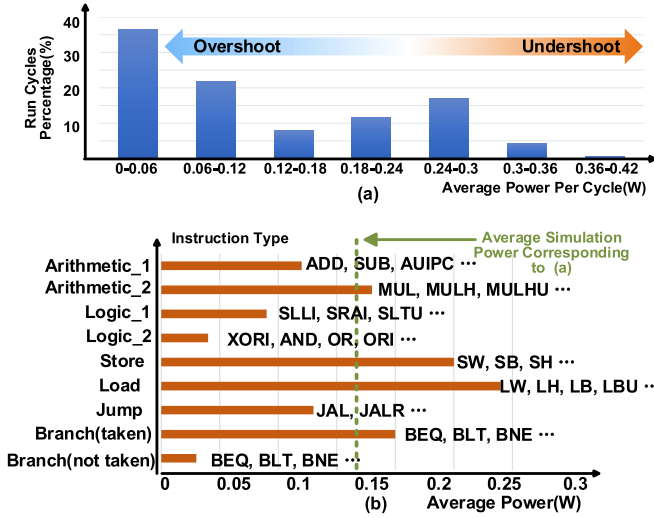


Fig. 4. (a) Histogram of cycle-wise dynamic power distribution of RISC-V CPU running benchmarks from Table I. (b) Relations between instructions and RISC-V CPU power.

stacked PMOS and NMOS transistors using core transistors as main power switches avoiding overstressing the devices. The spiral inductor has an inductance of around 3 nH, while the output capacitor is equivalent to the capacitance of the RISC-V digital core, which is around 2.1 nF according to pre-silicon extraction. There is no additional on-die decoupling capacitor. All control logics and drivers operate at 1.2 V regulated from an integrated LDO. An ac coupling capacitor is used to level shift the gate drive voltage for PMOS by ~ 0.9 V avoiding the expensive and slow level shifters. Comparators in the loop control module detect output voltage and generate the UP/DOWN signal to provide feedback to the PWM generation module. A divided-down clock at only 10–50 MHz is used as the loop clock for feedback control. For machine learning-based rapid regulation, a fast PWM modulation module (FPWMM) is implemented to receive the control signals issued by MLE one cycle earlier than the predicted event and modulate the switch control signals for the power cores in an average processing time of half-cycle.

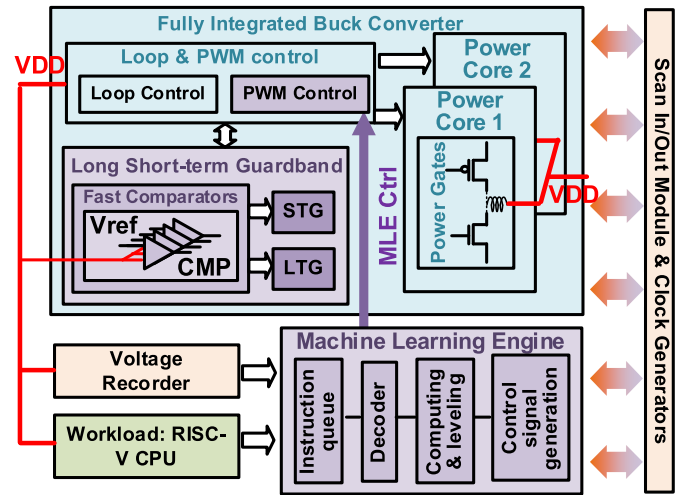


Fig. 5. Overview of the proposed architecture, including fully integrated buck converter, MLE, RISC-V CPU core, and voltage recorder.

C. Voltage Recorder

Even with a high-speed oscilloscope, it is nearly impossible to capture the accurate gigahertz-level supply waveforms from external measurements due to the packaging inductance and speed limitation of the probes. Hence, an on-chip VR is built to provide a more faithful observation of the chip's supply conditions. The VR consists of a fast comparator array and an on-chip memory. The detected voltage is encoded into a thermometer code and stored in memory, as shown in Fig. 6(b). To minimize the effect of the comparator offset, a post-silicon calibration was performed by directly connecting an external power supply to the CPU and providing fine-grained reference voltages to the comparator. This allows the comparator's offset to be measured and compensated for during real-time testing. It should be noted that this calibration method was specifically developed for testing and has certain limitations in capturing PVT variations. More advanced calibration techniques might be required in practice. The voltage detection resolution can be changed by repeating the same measurement with different threshold levels for the comparators. In our test chip, a 4-bit

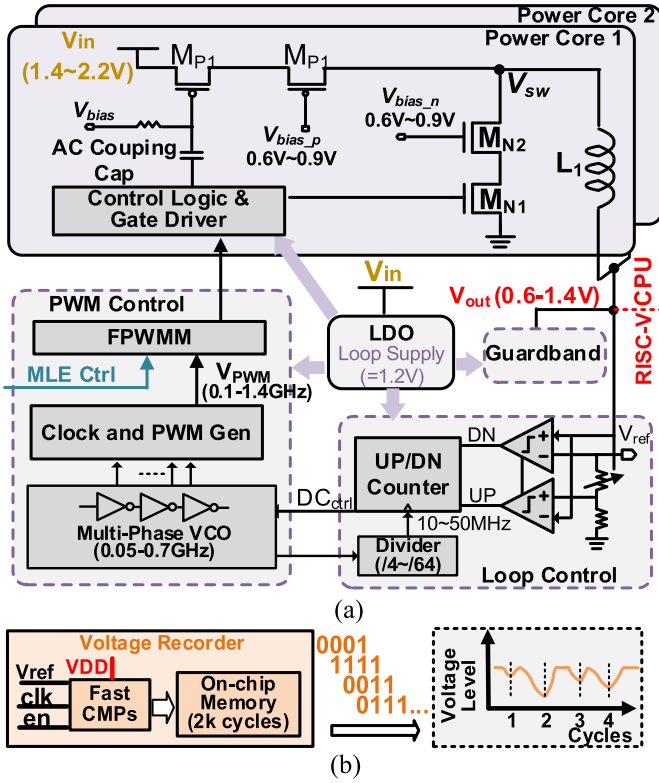


Fig. 6. (a) Structure of fully integrated buck converter. (b) On-chip voltage recorder.

thermometer coding with four reference levels is used with the highest resolution of 25 mV.

IV. MACHINE LEARNING ENGINE AND DROOP GUARDBAND

A. Model Analysis and Comparison

As discussed in Section II, both CPU information and real-time voltage are needed to perform a runtime droop prediction. All the benchmarks from Table I were simulated on RTL RISC-V CPU core to record the digital traces that are prepared for extracting power proxies, i.e., signals that are most relevant to power consumption within a certain time unit. Current/voltage waveforms corresponding to each program are generated from power analysis with mixed-signal simulation and further converted into cycle-wise discrete data. Runtime supply voltage is divided into three levels from the nominal voltage, i.e., undershoot, normal, and overshoot. A total of 320-bit raw power proxies from pipeline stages are dumped out based on RISC-V microarchitecture and used in droop prediction training with feature selection. More than 200k cycles are randomly selected from ten programs to cover a wide range of power scenarios. The training, validation, and testing datasets are split by a ratio of 7:1:2 for all the studied models. It should be noted that all the proxies are one-hot coded into single-bit features and can be decoded from the 32-bit instructions so that MLE only needs to access the 32-bit instructions from the instruction cache without intrusively accessing the internal signals from the CPU.

Two approaches are studied for droop prediction. The first one is single-stage prediction, which directly predicts the

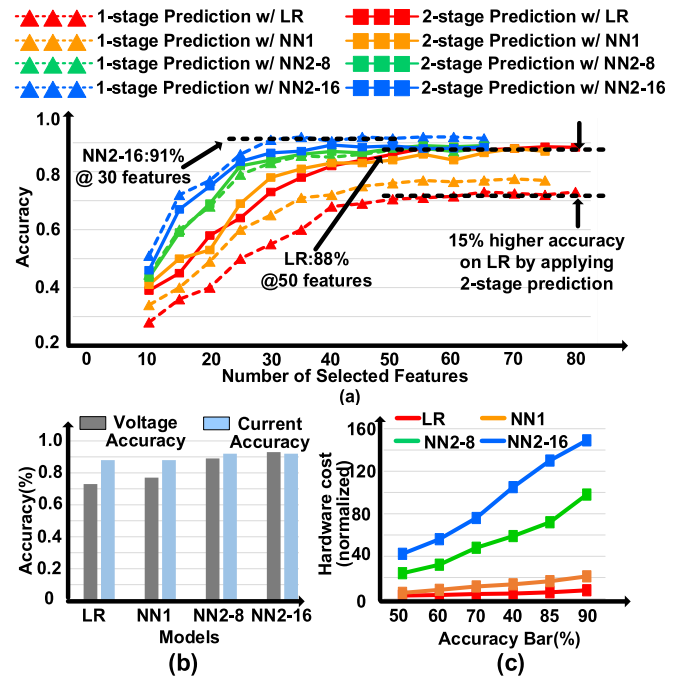


Fig. 7. (a) Accuracy of voltage prediction for four different models. (b) Prediction variation between current and voltage for different models. (c) Influence on accuracy by replacing the direct voltage prediction by current prediction with decision tree.

droop with input features and real-time voltage of previous cycles. Feature selection is performed with raw proxies among four different models, i.e., linear regression (LR), single-layer neural network (NN1), two-layer neural network with hidden layer size of 8 (NN2-8), and two-layer neural network with a hidden layer size of 16 (NN2-16). Accuracy comparison scaling with the number of most related features is plotted as dotted lines shown in Fig. 7(a). NN2-8 achieves $\sim 86\%$ accuracy with more than 35 input features, while NN2-16 achieves 91% accuracy with 30 input features. LR and NN1 can only achieve $\sim 72\%$ accuracy and $\sim 77\%$ accuracy due to the limited model complexity and the challenge of assembling information on voltage, current, and digital signals.

The second method is a two-stage prediction that divides the voltage prediction into two stages that first predict the current based on only the digital traces of the CPU and then realize the voltage prediction through a simplified, leveled prediction, e.g., decision tree, which incorporates both current prediction and transient supply voltage. The hardware cost of the decision tree is negligible. The elimination of voltage for the main ML model significantly eases the prediction challenges by allowing the model to only focus on the signal traces of the CPU. Fig. 7(b) shows the difference in maximum accuracy achieved by each model for current and voltage prediction. Around 5%–15% higher accuracy is observed on current prediction on LR, NN1, and NN2_8, while the NN2_16 model shows little difference in both current and voltage prediction due to larger model size. For the two-stage prediction, the ground truth data of real-time voltage and current are utilized to train a decision tree achieving 97% accuracy, which is further appended to the first-stage current prediction. As plotted with solid lines shown

in Fig. 7(a), by applying two-stage prediction, an accuracy improvement of 15% on LR and 11% on NN1 is observed. The accuracy change of NN2-8 is less than 1% on average. The accuracy of NN2-16 drops by $\sim 2\%$ when applying current prediction with the decision tree because the loss of the decision tree degrades the overall performance when a larger model of NN2-16 handles both voltage and current prediction similarly. As a result, two-stage prediction enables the LR to achieve an accuracy of 88% with around 50 input features, which is only 3% lower than the two-layer neural network using single-stage prediction.

As the larger model achieves higher accuracy but requires more hardware resources, a tradeoff between accuracy and hardware cost is crucial to the proactive PM scheme. A comparison of the synthesized numbers of logic gates for the models discussed above is shown in Fig. 7(c). To achieve accuracy higher than 85%, the LR consumes around $3\times$, $10\times$, and $20\times$ less than NN1, NN2-8, and NN2-16, respectively. Although a multi-layer neural network may be more beneficial when it comes to a larger system, given the simple RISC-V core test vehicle in this work, LR is chosen for implementation providing $3\times$ – $20\times$ hardware cost saving with less than 5% prediction accuracy loss. A total of 52-bit features are selected from the 320-bit raw power proxies for the on-chip MLE.

To provide better granularity and precision for implementing the regulation, undershoot is further divided into two levels, i.e., Undershoot_Weak and Undershoots_Strong, which are going to be regulated with different feedback strengths. Programs in Table I were tested for prediction on different levels. True positive rate (TPR) and false positive rate (FPR) are key indicators for droop prediction performance. TPR for undershoots/overshoots equals the number of correctly predicted undershoots/overshoots divided by the total number of real undershoots/overshoots. FPR for undershoots/overshoots equals the number of non-undershoots/non-overshoots incorrectly predicted as undershoots/overshoots divided by the total number of real non-undershoots/non-overshoots.

Variations can be observed among programs that prediction on strong undershoots is better than weak undershoots for some programs, while it is opposite for other programs. The overall TPR combining two levels is also shown in Fig. 8(a) as purple bars representing the overall undershoot prediction, which also corresponds to the original accuracy shown in Fig. 7(a). The TPR of overall undershoots is higher than the single-level undershoot prediction in most programs because some misprediction of Undershoot_Strong events are predicted to Undershoot_Weak, while some Undershoot_Weak events are mispredicted to Undershoot_Strong. These two cases are still considered to be correct in overall undershoot prediction and will still trigger regulation on the droop, but different regulation strengths. Measurement results in Section V will show that the regulation in most such cases is enough to avoid triggering guardband. Therefore, the prediction accuracy of the overall undershoots represents the performance of droop mitigation better than separated single-level prediction.

All the programs shown in Fig. 8(a) and (b) achieve more than 85% TPR and less than 5% FPR for the overall undershoots. As will be discussed in Section V, the impact

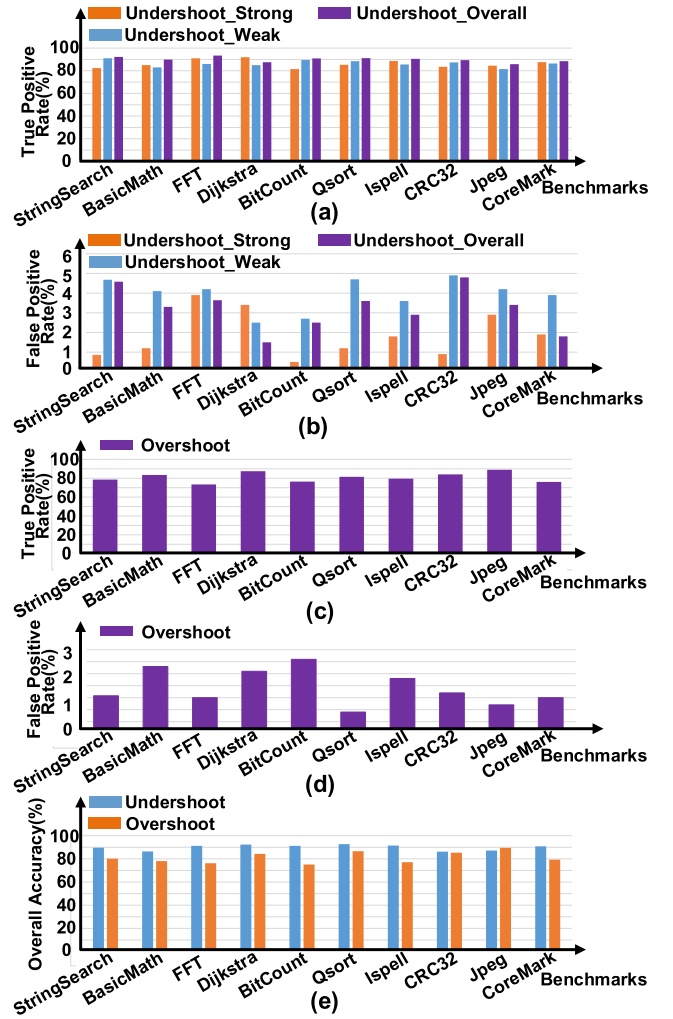


Fig. 8. (a) True positive rate for undershoots. (b) False positive rate for undershoots. (c) True positive rate for overshoots. (d) False positive rate for overshoots. (e) Overall accuracy for undershoots and overshoots.

on power efficiency at a TPR above 80% is negligible due to infrequent triggering of guardband operations.

An average TPR of around 80% and less than 5% FPR are achieved for overshoots as shown in Fig. 8(c) and (d), respectively.

The accuracies of undershoot and overshoot for each program are shown in Fig. 8(e) as a reference to demonstrate the overall prediction performance of each program.

B. Architecture and Dataflow of MLE

Based on the study in Section IV-A, a digital real-time MLE has been implemented near the microprocessor. Fig. 9 shows the overall architecture of the MLE. The fetched instructions from the RISC-V CPU are sent into the MLE 2–4 cycles before the corresponding power events, such as execution, memory, or write-back stages. Each instruction is stored in the instruction queue for three cycles to capture the impact of multiple instructions every cycle. A decoder implemented according to the RISC-V instruction set manual generates one-hot coded power proxies from 32-bit instructions. A total

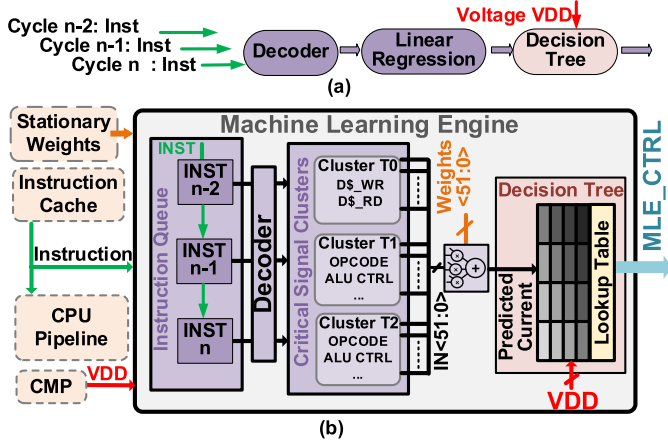


Fig. 9. (a) Simplified diagram showing MLE dataflow. (b) Architecture of MLE including the instruction queue, decoder, linear regression computation, and decision tree.

of 52-bit power proxies are further grouped into clusters of features, including Opcode, function code, ALU control, and memory access signals selected based on the study in previous sections. An LR unit takes the feature clusters as an input, and 4-bit stationary weights are applied to each bit of the 52-bit feature. As a result, only AND gates are needed, delivering multiplier-less operations, which reduces the cost. The regression results represent the upcoming workload current of the CPU and are then sent to the decision tree for voltage droop prediction. Incorporating the predicted workload current level and recorded real-time supply levels from the previous two cycles, the decision tree outputs the predicted droop and issues a predefined corresponding control command, which is sent to the buck converter every CPU cycle for power regulation. The receiver part as well as the regulation process will be introduced in Section IV-C.

C. FPWMM

The command issued by MLE is sent to FPWMM located in the PWM control module of the buck converter, as shown in Fig. 10. The original PWM switch signals travel through four delay paths with programmable delays. A pulse “stretcher” realized by an OR gate is used to increase the duty cycle of the PWM signal for reducing the undershoots. A pulse “trimmer” realized by an AND gate is used to reduce the duty cycle of the PWM signal to compensate for the overshoots. There are one “trimmer” and two “stretchers” with programmable delay to differentiate the regulation levels. The FPWMM maintains the original PWM signal if neither undershoots nor overshoots are predicted. The control command issued by the MLE controls which PWM signal to be sent to the power switches. In addition, a clock sampling circuit is used to synchronize the clock domain between MLE and the buck converter. As the “stretcher” and “trimmer” need to be switched back and forth across cycles and PWM comes with dynamic duty cycle, the control command must be converted to the buck converter clock domain before selecting PWM paths to avoid any glitches.

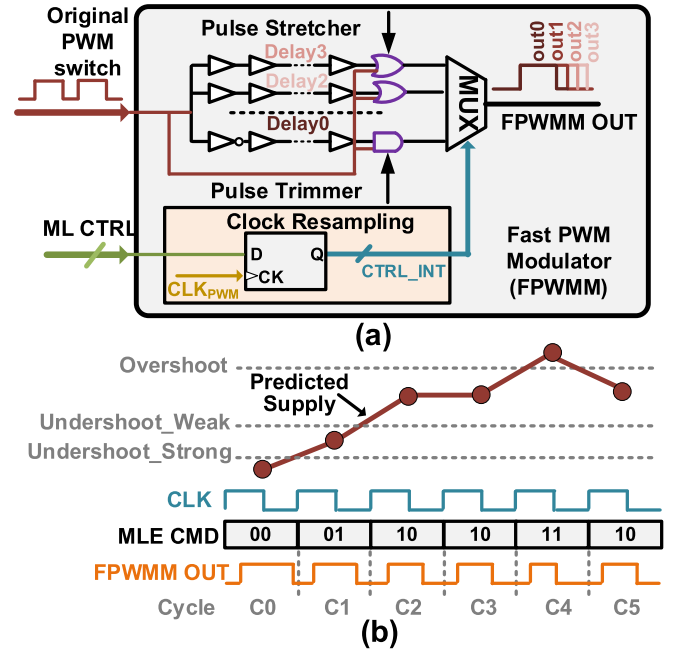


Fig. 10. (a) FPWMM circuits including pulse stretcher, pulse trimmer, and clock resampling module. (b) Examples of FPWMM response to predicted droop.

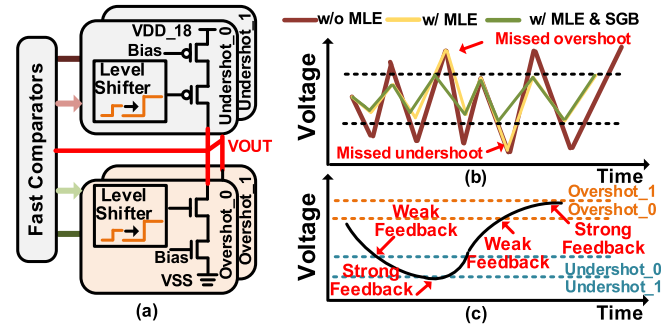


Fig. 11. (a) Structure of STG. (b) Example of STG compensation for MLE. (c) Feedback strength and thresholds.

D. Droop Guardband

To deal with the occasional misprediction of MLE and longer-term supply droop, which runs beyond the prediction time duration of MLE, two droop guardbands are implemented near the processor. The first one is STG, as shown in Fig. 11(a). A group of fast comparators is used to detect the real-time supply and control the power switches to bring the supply voltage within the required voltage range, as shown in Fig. 11(b). Multiple legs of power switches are deployed to deal with different droop levels based on the one-hot results from the comparators with different thresholds, as shown in Fig. 11(c). The STG can be activated within 300 ps. Because the STG behaves similar to a fast digital LDO, the STG should be activated as little as possible to reduce the efficiency loss. As machine learning prediction cannot achieve 100% accuracy, the impact of the execution of “safety net” needs to be considered.

The second submodule, i.e., LTG, is deployed to deal with the slower but larger power change, e.g., sudden CPU’s

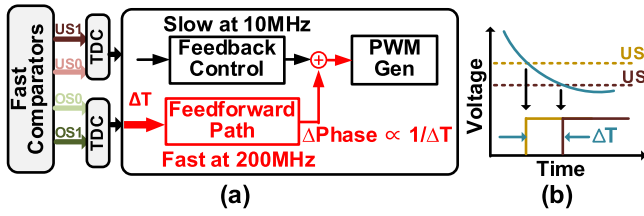


Fig. 12. (a) Structure of LTG. (b) Example of LTG dealing with long-term droop.

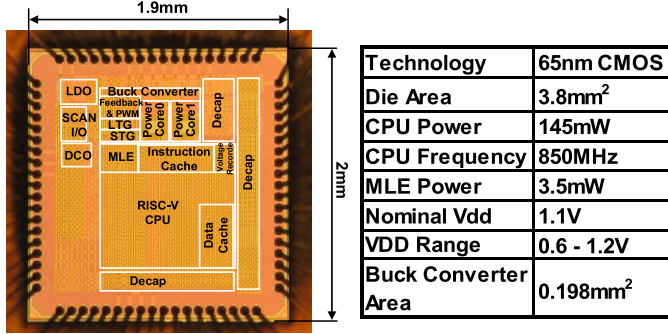


Fig. 13. Die photo and properties of 65-nm demonstrated chip.

wakeup or long-term supply noises due to PCB and package beyond the fine-grained regulation from our MLE predictions. In the LTG, additional PWM offset, which is inversely proportional to threshold crossing time, is issued through a fast feedforward path when supply droop is not recovered by ML/STG within two clock cycles. LTG reacts within ~ 5 CPU clock cycles, which is much faster than the main regulator loop at 10–50 MHz, as shown in Fig. 12(a).

V. SIMULATION AND MEASUREMENT RESULTS

A. Efficiency of Buck Converter Under Different Loads

A test chip was fabricated in a 65-nm process with a die photograph shown in Fig. 13. The RISC-V CPU operates at 850 MHz under a nominal voltage of 1 V with a power consumption of about 145 mW. MLE consumes 3.5-mW power, or about $\sim 2.4\%$ of CPU power, as an overhead.

Fig. 14 shows the measured efficiency of the buck converter on the demonstrated chip with varying output voltage and workload current. With an input voltage of 1.8 V and an output from 0.6 to 1.4 V, the peak efficiency is 79.3%. The voltage ripple of the buck converter ranges from 15 to 30 mV, depending on the operating switching frequency. The measured power consumption of RISC-V CPU is also shown in Fig. 14. It is worth mentioning that the efficiency of the implemented buck converter is relatively low due to the on-die spiral inductor. Its inductance is limited compared to air-core, magnetic-core, or off-chip inductors, requiring a higher switching frequency, which in turn leads to more switching loss and higher power consumption in control logics. In addition, the light CPU load does not reach the capacity of the proposed buck converter. As a result, the overall efficiency is relatively lower than other switching inductor-based power converters.

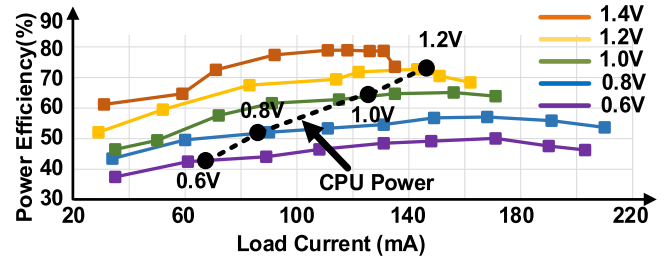


Fig. 14. Measured efficiency of buck converter scaling with output voltage and CPU workload current at input of 1.8 V.

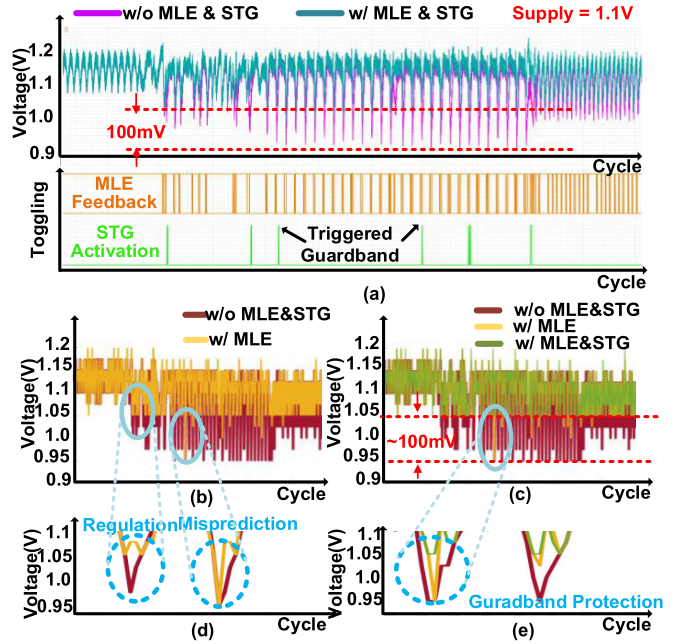


Fig. 15. Performance of MLE and STG verification with program StringSearch. (a) Simulated MLE and STG indicating the behavior of MLE and STG. (b) Measurement results with only MLE. (c) Measurement results with both MLE and STG. (d) Detailed view of MLE regulation and misprediction. (e) Detailed view of STG protection on MLE misprediction.

B. RISC-V CPU Supply Regulation Under MLE and Droop Guardband

The CPU functionality under dynamic supply droops was verified by running benchmark programs and scanning out all internal register files and caches. The VR can trace the real-time supply running on a program. Multiple benchmark programs have been run on the SoC with performance and ML accuracy evaluated. Figs. 15 and 16 show two examples of the measured supply waveforms for different benchmarks captured from the VR in comparison with the corresponding simulation results.

As shown in Fig. 15(a), a section of CPU operation from StringSearch is simulated showing that around 100-mV voltage droop is reduced by activating the MLE control and droop guardband. Measurements were performed on the same part of the program as the simulation. A small offset may be added to the threshold to account for possible variations between the real-time measurements and the simulation. The supply voltage stored in the VR is plotted in Fig. 15(b)–(e) and three modes of operations are recorded: 1) baseline without

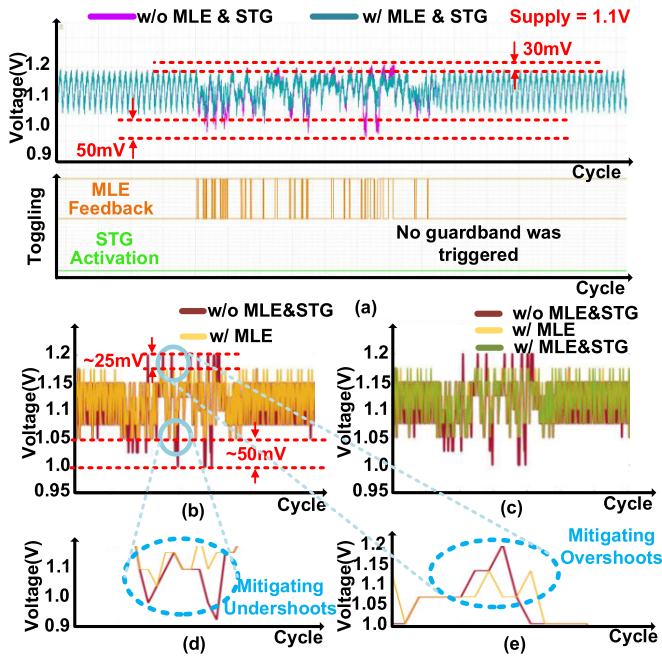


Fig. 16. Performance of MLE and STG verification with program CoreMark. (a) Simulated MLE and STG indicating the behavior of MLE and STG. (b) Measurement results with only MLE. (c) Measurement results with both MLE and STG. (d) Example of MLE mitigation on undershoots. (e) Example of MLE regulation on overshoots.

ML or STG; 2) operation with ML core but without STG; and 3) operation with both ML core and STG. With only MLE enabled, supply droop of more than 99% runtime is mitigated, while a few incidents remain below the threshold due to MLE misprediction or insufficient mitigation, as shown in Fig. 15(b) and (d). Such incidents count for less than 1% of program run cycles with negligible power loss. By activating STG, incidents are eliminated by the guardband, as shown in Fig. 15(c) and (e). A few overcompensations are observed due to the feedback from MLE or STG, while the amplitude of the noise is within the nominal voltage range. The frequency of such overcompensation is less than 1% of the total amount of recorded cycles with little influence on the overall performance. Overall, a droop reduction of ~ 100 mV is achieved when activating both MLE and STG.

Fig. 16 shows another example from the program CodeMark. In addition to the droop mitigation of around 50 mV observed from the VR, overshoots are also effectively removed by MLE, which provides another around 25-mV noise reduction for supply, as shown in Fig. 16(b). STG was not triggered in this test case since MLE predicted all the undershoots and overshoots correctly and provided enough feedback.

To verify LTG, VR was turned on as soon as the program was launched. With a sudden workload change, the supply dramatically dropped and lasted until the next power converter response. By applying LTG, a five-cycle recovering from a power surge of 5–150 mW is achieved, as shown in Fig. 17.

Three programs from Table I were tested on the chip, which are BM1: StringSearch, BM5: BasicMath, and BM10: CoreMark, and the average frequency is shown in Fig. 18(a). With activated MLE and STG, a frequency improvement of

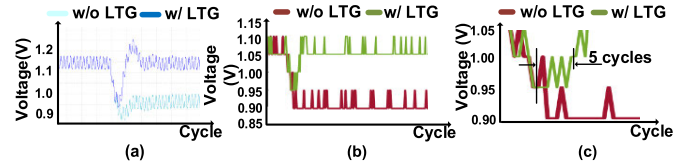


Fig. 17. (a) Simulated LTG response during CPU wakeup. (b) Measured LTG response to the same event. (c) Detailed response time for LTG.

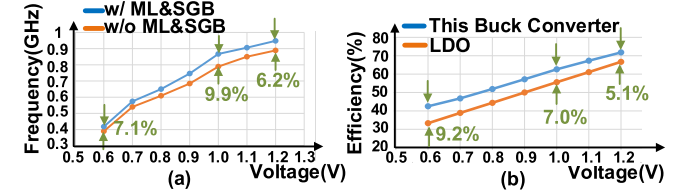


Fig. 18. (a) Frequency improvement with MLE and droop guardband measured with programs from Table I. (b) Efficiency improvement with MLE and droop Guardband with supply voltage scaling and comparison with conventional LDO.

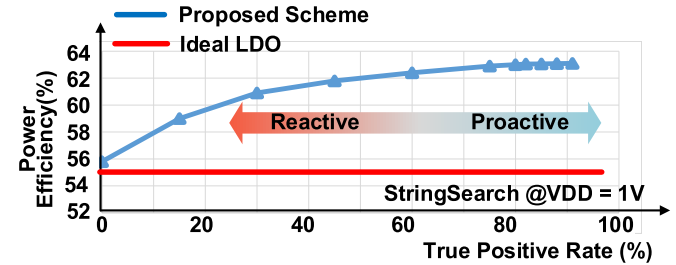


Fig. 19. Measured power efficiency of program StringSearch running on CPU with MLE accuracy scaling.

6.2%–9.9% was achieved with the supply voltage ranging from 0.6 to 1.2 V due to the supply margin saving, as shown in Figs. 15 and 16.

The proposed proactive PM scheme enables fast regulation from a buck converter, which is typically slow. Compared with digital LDO for providing similar cycle-level droop detection and mitigation using fast LDO methods [20], 5.1%–9.2% efficiency improvement was achieved with scaled supply voltage from 0.6 to 1.2 V, as shown in Fig. 18(b).

To analyze the impact of MLE accuracy on power efficiency, the partial program with high regulation activities from StringSearch was tested with ML models with different TPRs, which will lead to different activation rates of droop guardband. The measured power efficiency was recorded, as shown in Fig. 19. The degradation on efficiency due to the droop guardband is negligible when more than 80% TPR is achieved by MLE. As MLE TPR drops, the more frequent activation of droop guardband leads to efficiency loss due to the LDO-like charge injection and the efficiency declines to the level of an ideal LDO as the regulation from MLE is completely disabled, demonstrating the benefit of the proactive operations.

A comparison with prior droop mitigation schemes is shown in Table II. Compared with prior reactive techniques that directly detect the voltage droop, the MLE in the proposed proactive scheme enables a droop prediction in advance based on the behaviors of the microprocessor. To deal with the supply

TABLE II
COMPARISON WITH PRIOR WORKS

	ISSCC '21 [1]	ISSCC '22 [3]	ISSCC '21 [9]	JSSC '22 [14]	JSSC '11 [15]	VLSI '19 [19]	VLSI '19 [20]	ISSCC '18 [24]	ISSCC '21 [25]	ISSCC '20 [27]	JSSC '21 [28]	This work
Mitigation Scheme	Reactive						Proactive					Proactive
Process	7nm	5nm	7nm	65nm	45nm	22nm	65nm	14nm	7nm	14nm	7nm	65nm
Processor	CPU/GP U/APU	ARM CPU	5G Mobile SoC	N/A	Resilient microprocess or	N/A	N/A	Z14™	AI Chip	Z15	Hexagon™ DSP	RISC-V CPU
Droop Detection/Prediction	PVT Sensor	Peak Low Detector	N/A	Droop Detector	Tunable replica circuits	Event Detection and Sequence FSM	Domino Event Detection	Critical Path Sensor	Layer-based Workload Prediction	Critical Path Sensor	On-die Digital Power Meter	On-die Machine Learning Engine
Regulation Method	DVFS	AVS with FLL	Clock Gating	Timing Error Mitigation	Recovery Circuits	Computational LDO	LDO W/ PI Control	Instruction Throttling	Workload-aware Throttling	Throttle Actuation Circuits	Clock Gating	Fast PWM Modulation W/ Buck Converter
Vin (V)	N/A	N/A	N/A	N/A	N/A	0.55-1.2	0.5-1.0	N/A	N/A	N/A	N/A	1.8
Vout (V)	N/A	N/A	N/A	0.8-1.2	0.6-1	0.5-1.15	0.45-0.9	N/A	0.55-0.75	N/A	N/A	0.6-1.4
Operating Frequency	CPU: 2.6-2.8 GHz	CPU: 3.4GHz	CPU: 3.0GHz	Droop Detector: 500MHz	Processor: 1.5GHz	N/A	Detector: 250MHz	N/A	AI Chip: 1.0GHz	CPU: 5.2GHz z Control: 2.6GHz	N/A	CPU: 0.6-1.2GHz F _{sw} : 0.6-1GHz F _{loop} : 10-50MHz
Response Time	1ms	μs to ms	N/A	2-3.5ns	N/A	1ns	0.2ns	N/A	N/A	N/A	A few Clk Cycles	3~4 CPU Cycles for MLE
Equivalent Improvement	frequency	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.00%	9.90%
	Efficiency	13.00%	N/A	N/A	25.00%	N/A	N/A	N/A	N/A	N/A	N/A	9.20%
	Supply Reduction	N/A	4.00%	7.70%	12.00%	10.00%	N/A	N/A	N/A	N/A	5.00%	N/A

droop in modern SoCs that run at gigahertz, such prediction enables utilizing low-speed buck converter and achieves higher efficiency than conventional fast LDO. Compared with all the recent proactive schemes that utilize the throttling technique, by utilizing a fully integrated buck converter with FPWMM, this work can remove the droop without performance degradation. The proposed proactive PM scheme achieves 9.9% higher CPU frequency or equivalent 9.2% higher power efficiency. The response time of MLE is 3–4 cycles for MLE processing. The overhead of the MLE is about 2.4% of CPU power and 3.9% of CPU area, while the overhead of the additional regulation circuit in buck converter is negligible.

VI. CONCLUSION

In this article, a proactive PM scheme is presented with machine learning-based droop prediction and fast guardband on the RISC-V microprocessor. By responding in advance before the droop events, the efficiency benefit of a fully integrated buck converter is exploited. FPWMM directly regulating the supply voltage eliminates the need of throttling in common proactive schemes that degrade the system performance. Demonstrated on a 65-nm test chip, the proposed scheme resolves the speed and efficiency tradeoff between conventional power converters and LDOs achieving a 9.2% efficiency improvement.

REFERENCES

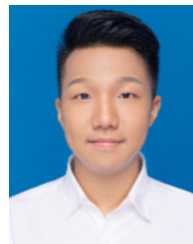
- [1] B.-J. Huang et al., “35.1 An octa-core 2.8/2 GHz dual-gear sensor-assisted high-speed and power-efficient CPU in 7 nm FinFET 5G smartphone SoC,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2021, pp. 490–492.
- [2] H. Mair et al., “2.5 A 7 nm FinFET 2.5 GHz/2.0 GHz dual-gear octa-core CPU subsystem with power/performance enhancements for a fully integrated 5G smartphone SoC,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2020, pp. 50–52.
- [3] A. Nayak et al., “A 5 nm 3.4 GHz tri-gear ARMv9 CPU subsystem in a fully integrated 5G flagship mobile SoC,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2022, pp. 50–52.
- [4] M. Cho et al., “Postsilicon voltage guard-band reduction in a 22 nm graphics execution core using adaptive voltage scaling and dynamic power gating,” *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 50–63, Jan. 2017.
- [5] P. Mosalikanti et al., “29.3 80 ns fast-lock 0.4-to-6.5 GHz clock generator with self-referenced asynchronous adaptive droop mitigation,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2021, pp. 408–410.
- [6] T. Jia, R. Joseph, and J. Gu, “An instruction driven adaptive clock phase scaling with timing encoding and online instruction calibration for a low power microprocessor,” in *Proc. IEEE 44th Eur. Solid State Circuits Conf. (ESSCIRC)*, Dresden, Germany, Sep. 2018, pp. 94–97.
- [7] T. Jia, R. Joseph, and J. Gu, “An adaptive clock management scheme exploiting instruction-based dynamic timing slack for a general-purpose graphics processor unit with deep pipeline and out-of-order execution,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2019, pp. 318–320.
- [8] T. Jia, Y. Ju, and J. Gu, “A compute-adaptive elastic clock-chain technique with dynamic timing enhancement for 2D PE-array-based accelerators,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2020, pp. 482–484.
- [9] H. Chen et al., “4.1 A 7 nm 5G mobile SoC featuring a 3.0 GHz tri-gear application processor subsystem,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2021, pp. 54–56.
- [10] K. A. Bowman, C. Tokunaga, T. Karnik, V. K. De, and J. W. Tschanz, “A 22 nm dynamically adaptive clock distribution for voltage droop tolerance,” in *Proc. Symp. VLSI Circuits (VLSIC)*, Honolulu, HI, USA, Jun. 2012, pp. 94–95.
- [11] K. A. Bowman et al., “A 16 nm auto-calibrating dynamically adaptive clock distribution for maximizing supply-voltage-droop tolerance across a wide operating range,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2015, pp. 1–3.
- [12] M. S. Floyd et al., “26.5 adaptive clocking in the POWER9™ processor for voltage droop protection,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2017, pp. 444–445.
- [13] J. Tschanz et al., “Adaptive frequency and biasing techniques for tolerance to dynamic temperature-voltage variations and aging,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2007, pp. 292–604.

- [14] Y. Shifman, I. Stanger, N. Shavit, R. Taco, A. Fish, and J. Shor, "A method for mitigation of droop timing errors including a 500 MHz droop detector and dual mode logic," *IEEE J. Solid-State Circuits*, vol. 57, no. 2, pp. 596–608, Feb. 2022.
- [15] K. A. Bowman et al., "A 45 nm resilient microprocessor core for dynamic variation tolerance," *IEEE J. Solid-State Circuits*, vol. 46, no. 1, pp. 194–208, Jan. 2011.
- [16] R. Muthukaruppan et al., "A digitally controlled linear regulator for per-core wide-range DVFS of atom™ cores in 14 nm tri-gate CMOS featuring non-linear control, adaptive gain and code roaming," in *Proc. 43rd IEEE Eur. Solid State Circuits Conf.*, Leuven, Belgium, Sep. 2017, pp. 275–278.
- [17] M. Zelickson et al., "14.3 A digital low-dropout (LDO) linear regulator with adaptive transfer function featuring 125A/mm² power density and autonomous bypass mode," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2023, pp. 230–232.
- [18] D. Kim, S. Kim, M. Seok, H. Ham, and J. Kim, "0.5 V-VIN, 165-mA/mm² fully-integrated digital LDO based on event-driven self-triggering control," in *Proc. IEEE Symp. VLSI Circuits*, Honolulu, HI, USA, Jun. 2018, pp. 109–110.
- [19] Z. K. Ahmed et al., "A variation-adaptive integrated computational digital LDO in 22 nm CMOS with fast transient response," in *Proc. Symp. VLSI Circuits*, Kyoto, Japan, Jun. 2019, pp. C124–C125.
- [20] S. J. Kim, D. Kim, Y. Pu, C. Shi, and M. Seok, "A 0.5–1 V input event-driven multiple digital low-dropout-regulator system for supporting a large digital load," in *Proc. Symp. VLSI Circuits*, Kyoto, Japan, Jun. 2019, pp. C128–C129.
- [21] W. Kim, D. M. Brooks, and G.-Y. Wei, "A fully-integrated 3-level DC/DC converter for nanosecond-scale DVS with fast shunt regulation," in *Proc. IEEE Int. Solid-State Circuits Conf.*, San Francisco, CA, USA, Feb. 2011, pp. 268–270.
- [22] P. Kumar et al., "A 0.4 V 1 V 0.2A/mm² 70% efficient 500 MHz fully integrated digitally controlled 3-level buck voltage regulator with on-die high density MIM capacitor in 22 nm tri-gate CMOS," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, San Jose, CA, USA, Sep. 2015, pp. 1–4.
- [23] X. Chen et al., "A 65 nm fully-integrated fast-switching buck converter with resonant gate drive and automatic tracking," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, San Antonio, TX, USA, Apr. 2023, pp. 1–2.
- [24] C. Vezirtzis et al., "Droop mitigation using critical-path sensors and an on-chip distributed power supply estimation engine in the z14™ enterprise processor," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2018, pp. 300–302.
- [25] A. Agrawal et al., "9.1 A 7 nm 4-core AI chip with 25.6 TFLOPS hybrid FP8 training, 102.4 TOPS INT4 inference and workload-aware throttling," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2021, pp. 144–146.
- [26] V. J. Reddi, M. Gupta, G. Holloway, M. D. Smith, G.-Y. Wei, and D. Brooks, "Predicting voltage droops using recurring program and microarchitectural event activity," *IEEE Micro*, vol. 30, no. 1, p. 110, Jan. 2010.
- [27] C. Berry et al., "2.7 IBM z15: A 12-core 5.2 GHz microprocessor," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2020, pp. 54–56.
- [28] V. K. Kalyanam, E. Mahurin, K. A. Bowman, and J. A. Abraham, "A proactive voltage-droop-mitigation system in a 7 nm Hexagon™ processor," *IEEE J. Solid-State Circuits*, vol. 56, no. 4, pp. 1166–1175, Apr. 2021.
- [29] A. Drake et al., "A distributed critical-path timing monitor for a 65 nm high-performance microprocessor," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2007, pp. 398–399.
- [30] D. Wolpert et al., "Cores, cache, content, and characterization: IBM's second generation 14-nm product, z15," *IEEE J. Solid-State Circuits*, vol. 56, no. 1, pp. 98–111, Jan. 2021.
- [31] Z. Xie et al., "APOLLO: An automated power modeling framework for runtime power introspection in high-volume commercial microprocessors," in *Proc. 54th Annu. IEEE/ACM Int. Symp. Microarchitecture (MICRO)*, Oct. 2021, pp. 1–14.
- [32] S. B. Nasir and A. Raychowdhury, "All-digital linear regulators with proactive and reactive gain-boosting for supply droop mitigation in digital load circuits," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Montreal, QC, Canada, May 2016, pp. 205–208.
- [33] F. Atallah et al., "19.3 A 7 nm all-digital unified voltage and frequency regulator based on a high-bandwidth 2-phase buck converter with package inductors," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, San Francisco, CA, USA, Feb. 2019, pp. 316–318.
- [34] S. Bang et al., "An all-digital, V_{MAX}-compliant, stable, and scalable distributed charge injection scheme in 10-nm CMOS for fast and local mitigation of voltage droop," *IEEE J. Solid-State Circuits*, vol. 55, no. 7, pp. 1898–1908, Jul. 2020.
- [35] G. Contrera, and M. Martonosi, "Power prediction for Intel XScale™ processors using performance monitoring unit events," in *Proc. Int. Symp. Low Power Electron. Design (ISLPED)*, San Diego, CA, USA, 2005, pp. 221–226.
- [36] X. Chen et al., "Proactive power regulation with real-time prediction and fast response guardband for fine-grained dynamic voltage droop mitigation on digital SoCs," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Kyoto, Japan, Jun. 2023, pp. 1–2.
- [37] M. R. Guthaus, J. S. Ringenberg, D. Ernst, T. M. Austin, T. Mudge, and R. B. Brown, "MiBench: A free, commercially representative embedded benchmark suite," in *Proc. 4th Annu. IEEE Int. Workshop Workload Characterization*, Austin, TX, USA, Dec. 2001, pp. 3–14.
- [38] T. Jia and J. Gu, "A 0.3–0.86 V fully integrated buck regulator with 2 GHz resonant switching for ultra-low power applications," in *Proc. Symp. VLSI Circuits*, Kyoto, Japan, Jun. 2017, pp. C208–C209.
- [39] S. S. Amin and P. P. Mercier, "A fully integrated Li-ion-compatible hybrid four-level DC–DC converter in 28-nm FDSOI," *IEEE J. Solid-State Circuits*, vol. 54, no. 3, pp. 720–732, Mar. 2019.
- [40] N. Tang, W. Hong, B. Nguyen, Z. Zhou, J.-H. Kim, and D. Heo, "Fully integrated switched-inductor-capacitor voltage regulator with 0.82-A/mm² peak current density and 78% peak power efficiency," *IEEE J. Solid-State Circuits*, vol. 56, no. 6, pp. 1805–1815, Jun. 2021.



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